

Dear Editor and Reviewers,

Thank you very much for the thorough and constructive feedback on our manuscript. We are grateful for the time and care all three reviewers invested, and we are pleased that the paper was well received overall. We have revised the manuscript to address every point raised by Reviewers 1 and 2, including a clearer explanation of the roadmap's sequencing logic, a stronger theoretical grounding in sustainability principles, explicit flagging of generative AI tool use, a new figure visualizing the roadmap, a sharpened description of our qualitative coding procedure, and a consistent framing of our study as an exploratory pilot throughout the manuscript. We also carried out a full language pass to address the phrasing issues noted by Reviewer 2.

We thank Reviewer 3 for the positive assessment of the paper's practical framing and discussion; while no specific changes were requested, several of the revisions made in response to Reviewers 1 and 2 further reinforce the strengths highlighted in that review.

A detailed, point-by-point account of all changes is provided in the following Response to Reviewers document. We believe these revisions have substantially strengthened the manuscript, and we thank the reviewers again for their valuable input.

Sincerely, The Authors

Response to Reviewers

Reviewer 1 (Accept, Language: good)

R1, Item 1 (Sequenzierung)

Comment: The reviewer noted that the ordering of concepts and tools within the roadmap sometimes appears unclear, and asked us to review the structure/logic of the sequencing and justify it more explicitly.

Change made: New Section 2.3 added, explaining both the inter-chapter progression (grounded in Bloom's Taxonomy) and the intended intra-chapter ordering logic (standards → tools → patterns → resources), with a forward-reference to the workshop findings on where this logic was not fully transparent to participants.

R1, Item 2 (Kontextualisierung)

Comment: The reviewer noted that foundational sustainability principles are not sufficiently situated within the paper, and asked for a stronger theoretical/conceptual embedding.

Change made: New opening paragraph added to Section 2, grounding the roadmap's "Understanding the Problem" phase in established sustainability theory (planetary boundaries [Ro09], triple bottom line [El97]) and linking it to the ICT-specific framings already used in the roadmap.

R1, Item 3 (Kennzeichnung von KI-Tools)

Comment: The reviewer asked that tools such as NotebookLM, which may introduce bias, be explicitly flagged as requiring particular attention.

Change made: Section 3.2 revised: NotebookLM is now explicitly flagged as a generative AI tool that can introduce stylistic/representational bias, with a clarification that its use was scoped to illustrative imagery only and did not influence the roadmap's conceptual or textual content.

R1, Item 4 (Trade-off-Darstellung)

Comment: The reviewer asked us to more strongly work out measurement and decision trade-offs, for instance through comparison tables or scenario analyses, to give developers a more robust basis for decision-making.

Change made: New paragraph added to the Discussion (Sec. 5), proposing comparative metric overviews and scenario-style framings as concrete mechanisms for surfacing trade-offs, explicitly grounded in participants' PUE critique (Sec. 4.3) and flagged as future work beyond the current pilot's scope.

R1, Item 5 (Provenienz/wissenschaftliche Fundierung)

Comment: The reviewer asked us to more transparently present the provenance of the roadmap and the scientific foundations underlying it.

Change made: New closing paragraph added to Section 2.2, making explicit how the roadmap's content was derived from established reference frameworks/standards.

R1, Item 6 (Testfälle/konkrete Beispiele)

Comment: Given the heterogeneity of software projects, the reviewer suggested trialing the roadmap on concrete examples/test cases, at least as an outlook, and potentially relating this to agentic AI as an emerging paradigm.

Change made: Section 6 extended with one sentence committing to trialing the roadmap against concrete test cases across heterogeneous project types.

Reviewer 2 (Strong Accept, Language: good – “some phrases have to be corrected”)

R2, Item 1 (Forschungslücke & Beitrag)

Comment: The reviewer asked that the scientific gap, research goal, and contribution of the paper be formulated more explicitly in the introduction.

Change made: One sentence added in Section 1, explicitly stating the research gap (limited empirical evidence on developer perception/adoption of roadmap-based green coding training) and this paper's contribution (participatory evaluation of a concrete roadmap artefact), placed just before the roadmap.sh description.

R2, Item 2 (Sprachliche Korrekturen)

Comment: The reviewer noted that the language is generally good but that some phrases require correction, specifically flagging “where in the part of the HTW Berlin was shown.”

Change made: The flagged sentence was rewritten for clarity (“...in which the HTW Berlin team demonstrated that green coding can be integrated into curricula...”). We also corrected a rendering artifact (“Intervention and Training is structured” → “‘Intervention and Training’ is structured”). A full language pass of the manuscript was conducted following all content revisions to address the reviewer's general request for language correction.

R2, Item 3 (Visuelle Roadmap-Darstellung)

Comment: The reviewer noted the absence of a figure or screenshot of the roadmap and asked for a compact graphic showing phases, chapters, and the relationships between nodes, tools, and outcomes.

Change made: Added Figure 1: a legible screenshot excerpt from the live roadmap.sh page, showing the “Measure & Assess” phase and its nodes (including Software Carbon Intensity), referenced in Section 2.2. Full-resolution screenshots of the complete roadmap are available via the Zenodo dataset [JKW26] and the roadmap.sh link [ro26] already cited in the paper.

R2, Item 4 (Liste der Roadmap-Themen kürzen)

Comment: The reviewer noted that the list of roadmap themes demonstrates depth but should be compressed and streamlined.

Change made: The bullet list in Section 2.2 was condensed, reducing each category from up to seven items to a representative four, while retaining all eight thematic categories.

R2, Item 5 (Methodik der qualitativen Inhaltsanalyse)

Comment: The reviewer asked us to sharpen the description of the qualitative content analysis, specifically the coding procedure, theme formation, the researchers involved, and the validation of interpretations.

Change made: Section 3.4 was revised to specify that one researcher conducted an initial inductive, open coding of the worksheets and plenary notes, without a predefined coding scheme. The resulting codes were then reviewed and iteratively grouped into higher-level themes through joint discussion among all authors, with disagreements resolved by consensus.

R2, Item 6 (Exploratives Framing der Studie)

Comment: Given the small sample of twelve participants, ten of them students, the reviewer asked us to frame the study as an exploratory evaluation rather than as general evidence of developer acceptance, optionally supported by additional references.

Change made: The study is now explicitly framed as an exploratory pilot at four points in the manuscript: the Abstract, the Introduction's contribution statement, Section 3.1, and the Conclusion, replacing stronger claims (e.g., “the results show”) with appropriately hedged language (“indicate,” “exploratory pilot”).

R2, Item 7 (Trennung Beobachtung vs. generelle Aussage)

Comment: The reviewer asked us to more clearly distinguish, in the Discussion, between actual workshop observations and general statements about green coding education.

Change made: The second paragraph of the Discussion (Section 5) was revised to explicitly attribute design implications to participant feedback rather than presenting them as standalone general claims (e.g., “one implication is” → “participants' feedback suggests”; “narrative framing is as important as...” → “narrative framing was perceived as being as important as..., at least for this group”). The trade-off-related implication is now explicitly flagged as a design hypothesis motivated by the findings, rather than a tested effect.

Location: Section 5, Discussion, second paragraph.

Reviewer 3 (Strong Accept, Language: good)

Comment: The reviewer gave a purely positive assessment, with no requested revisions, highlighting the workshop documentation, the practical application of green coding concepts, the connection between green coding and broader sustainability, and the Discussion and Conclusion sections.

We thank Reviewer 3 for their positive assessment of the paper. While Reviewer 3 did not request specific changes, we note that several revisions made in response to Reviewers 1 and 2 also strengthen the aspects Reviewer 3 highlighted:

- The new Figure 1 (a roadmap.sh screenshot excerpt of the “Measure & Assess” phase) grounds the paper's application-oriented framing in a concrete visual artefact, reinforcing the practical, applied character of the work noted by Reviewer 3.

- The revised Discussion (Section 5) more clearly distinguishes participant observations from broader design implications, which we believe further strengthens the analytical clarity that Reviewer 3 highlighted as a strength.
- The added paragraph on trade-off mechanisms extends the connection between green coding practice and systemic sustainability thinking that Reviewer 3 praised in the Discussion and Conclusion.

No manuscript changes were made specifically in response to Reviewer 3, as the review did not identify required revisions.